SPRINGFIELD GENERAL HOSPITAL

LABORATORY & DIAGNOSTIC REPORT

Patient Name: Robert Thompson **Date of Birth:** 03/15/1978 **MRN:** MED-2024-5567 **Account Number:** ACCT-SGH-44821 **Ordering Physician:** Dr. Sarah Williams, MD — Neurology **Collection Date:** January 8, 2024 **Report Date:** January 9, 2024

COMPLETE BLOOD COUNT (CBC) WITH DIFFERENTIAL

Specimen: Whole blood, EDTA **Collection Time:** 15:20

Test	Result	Units	Reference Range	Flag
WBC	7.2	x10³/µL	4.0–11.0	
RBC	5.12	x10⁶/µL	4.50–5.50	
Hemoglobin	14.8	g/dL	13.5–17.5	
Hematocrit	44.2	%	38.0–50.0	
MCV	86.3	fL	80.0–100.0	
MCH	28.9	pg	27.0–33.0	
MCHC	33.5	g/dL	32.0–36.0	
RDW	13.1	%	11.5–14.5	
Platelet Count	245	x10³/µL	150–400	
MPV	9.8	fL	7.5–11.5	

White Blood Cell Differential

Cell Type	Result (%)	Result (Abs)	Reference (%)	Reference (Abs)
Neutrophils	62.4	4.49	40–70	1.8–7.7
Lymphocytes	28.1	2.02	20–45	1.0–4.8
Monocytes	6.8	0.49	2–10	0.1–1.0
Eosinophils	2.1	0.15	1–6	0.0–0.5
Basophils	0.6	0.04	0–2	0.0–0.2

Interpretation: All values within normal limits. No evidence of infection, anemia, or hematologic abnormality.

BASIC METABOLIC PANEL (BMP)

Specimen: Serum **Collection Time:** 15:20

Test	Result	Units	Reference Range	Flag
Sodium	140	mEq/L	136–145	
Potassium	4.1	mEq/L	3.5–5.1	
Chloride	103	mEq/L	98–106	
CO2 (Bicarbonate)	25	mEq/L	22–29	
BUN	16	mg/dL	7–20	
Creatinine	0.9	mg/dL	0.7–1.3	
eGFR	>90	mL/min/1.73m²	>60	
Glucose	102	mg/dL	70–100	**H**
Calcium	9.4	mg/dL	8.5–10.5	
Anion Gap	12	mEq/L	8–16	

Interpretation: Glucose mildly elevated at 102 mg/dL (fasting status unknown — patient was in ED; likely non-fasting specimen). Renal function normal. Electrolytes within normal limits.

INFLAMMATORY MARKERS

Specimen: Serum **Collection Time:** 15:20

Test	Result	Units	Reference Range	Flag
ESR (Sed Rate)	12	mm/hr	0–15	
C-Reactive Protein (CRP)	0.8	mg/dL	0.0–1.0	

Interpretation: Inflammatory markers within normal limits. No evidence of systemic inflammatory process, infection, or malignancy.

COAGULATION STUDIES

Specimen: Citrated plasma **Collection Time:** 15:25

Test	Result	Units	Reference Range	Flag
Prothrombin Time (PT)	11.8	seconds	9.5–13.5	
INR	1.0	ratio	0.9–1.1	
Partial Thromboplastin Time (PTT)	28.4	seconds	25.0–35.0	

Interpretation: Coagulation profile normal. Patient cleared for epidural injection procedures.

URINALYSIS

Specimen: Clean catch, midstream **Collection Time:** 15:45

Test	Result	Reference
Color	Yellow	Yellow–Amber
Clarity	Clear	Clear
Specific Gravity	1.018	1.005–1.030
pH	6.0	5.0–8.0
Protein	Negative	Negative
Glucose	Negative	Negative
Ketones	Negative	Negative
Blood	Negative	Negative
Bilirubin	Negative	Negative
Urobilinogen	Normal	Normal
Nitrite	Negative	Negative
Leukocyte Esterase	Negative	Negative
WBC	0–2 /hpf	0–5 /hpf
RBC	0–1 /hpf	0–3 /hpf
Bacteria	None seen	None
Casts	None seen	None

Interpretation: Urinalysis within normal limits. No evidence of urinary tract infection, hematuria, or proteinuria.

ELECTROCARDIOGRAM (EKG)

Date: January 8, 2024 **Time:** 16:00 **Ordering Physician:** Dr. Michael Chen, MD **Interpreting Physician:** Dr. Rachel Kim, MD — Cardiology

Technical Details:

- 12-lead standard EKG
- Heart Rate: 78 bpm
- PR Interval: 162 ms
- QRS Duration: 88 ms

- QT/QTc: 384/438 ms

Findings:

- Normal sinus rhythm at 78 beats per minute
- Normal axis
- Normal P-wave morphology
- PR interval within normal limits
- QRS duration normal, no bundle branch block
- QTc interval normal
- No ST segment elevation or depression
- No T-wave inversions
- No pathological Q waves
- No chamber enlargement by voltage criteria

Interpretation: Normal EKG. No acute ischemic changes. No arrhythmia.

CHEST X-RAY (PORTABLE)

Date: January 8, 2024 **Time:** 16:15 **Ordering Physician:** Dr. Michael Chen, MD **Interpreting Physician:** Dr. Laura Bennett, MD — Radiology

Clinical Indication: Pre-operative evaluation; lower back pain, rule out thoracic pathology.

Technique: Single AP portable view.

Findings:

- Heart size normal
- Mediastinal contour normal
- Lungs are clear bilaterally — no infiltrate, consolidation, effusion, or pneumothorax
- Costophrenic angles are sharp
- No rib fractures identified
- Soft tissues unremarkable
- No free air under diaphragm

Impression: Normal portable chest X-ray. No acute cardiopulmonary process.

MRI LUMBAR SPINE WITHOUT CONTRAST

Date: January 8, 2024 **Time:** 17:30 **Ordering Physician:** Dr. Sarah Williams, MD **Interpreting Physician:** Dr. Laura Bennett, MD — Radiology **MRI Scanner:** 1.5T Siemens Magnetom **Sequences:** Sagittal T1, Sagittal T2, Axial T2 at L3-L4, L4-L5, L5-S1

Clinical Indication: Lower back pain with left lower extremity radiculopathy. Rule out disc herniation or spinal stenosis.

Findings:

L1-L2: Normal disc height and signal. No herniation or stenosis. Facet joints normal.

L2-L3: Normal disc height and signal. No herniation or stenosis. Facet joints normal.

L3-L4: Mild disc desiccation (T2 signal loss). Small broad-based disc bulge without significant canal narrowing. Mild bilateral facet arthropathy. Neural foramina patent bilaterally. No nerve root compression.

L4-L5: Moderate disc desiccation. **Left paracentral disc herniation measuring approximately 7mm AP x 12mm transverse.** The herniated disc material contacts and displaces the left L5 nerve root laterally and posteriorly. Mild narrowing of the left lateral recess. Right lateral recess patent. Central canal AP diameter reduced to 10mm (normal >12mm). Mild bilateral facet arthropathy with small joint effusions.

L5-S1: Normal disc height with mild desiccation. No herniation. Neural foramina patent bilaterally.

Conus medullaris: Normal in position (terminating at L1), signal, and morphology.

Vertebral bodies: Normal height and alignment. No fracture or malalignment. No suspicious marrow signal abnormality.

Paraspinal soft tissues: Mild edema of left paraspinal muscles at L4-L5 level.

Impression:

1. **Moderate left paracentral disc herniation at L4-L5 with compression of the left L5 nerve root.** Correlates with clinical presentation of left L5 radiculopathy.
2. Mild degenerative disc disease at L3-L4 without significant nerve root compression.
3. No spinal stenosis. No fracture or malalignment.

EMG / NERVE CONDUCTION STUDY

Date: January 9, 2024 **Performing Physician:** Dr. Sarah Williams, MD — Neurology **Clinical Indication:** Left lower extremity weakness and numbness. Evaluate for radiculopathy.

Nerve Conduction Studies

Motor Nerve Conduction:

Nerve	Stimulation Site	Recording Site	Latency (ms)	Amplitude (mV)	Velocity (m/s)	Normal
L Peroneal	Ankle	EDB	4.2	4.8	—	Lat <6.1, Amp >2.0
L Peroneal	Below fibula	EDB	10.8	4.5	48	Vel >40
L Tibial	Ankle	AH	4.0	8.2	—	Lat <6.0, Amp >3.0
L Tibial	Popliteal fossa	AH	12.4	7.8	46	Vel >40

Sensory Nerve Conduction:

Nerve	Stimulation Site	Recording Site	Latency (ms)	Amplitude (µV)	Velocity (m/s)	Normal
L Sural	Calf	Ankle	3.2	14.5	44	Lat <4.2, Amp >6.0, Vel >40
L Superficial Peroneal	Lower leg	Ankle	3.0	12.8	46	Lat <4.0, Amp >6.0, Vel >40

Interpretation: All motor and sensory nerve conductions within normal limits. No evidence of peripheral neuropathy or entrapment neuropathy.

Needle Electromyography

Muscle	Nerve/Root	Insertional	Fibs	PSW	Fascics	Recruitment	MUP
L Tibialis Anterior	Deep Peroneal / L4-L5	Increased	2+	2+	None	Reduced	Normal
L Ext Hallucis Longus	Deep Peroneal / L5	Increased	2+	2+	None	Reduced	Normal
L Peroneus Longus	Superficial Peroneal / L5	Increased	1+	2+	None	Mildly Reduced	Normal
L Med Gastrocnemius	Tibial / S1-S2	Normal	None	None	None	Normal	Normal
L Vastus Medialis	Femoral / L2-L4	Normal	None	None	None	Normal	Normal
L Tensor Fascia Lata	Superior Gluteal / L4-L5	Normal	None	None	None	Normal	Normal
L Lumbar Paraspinals (L4-L5)	Posterior Rami / L4-L5	Increased	1+	1+	None	—	—
R Tibialis Anterior	Deep Peroneal / L4-L5	Normal	None	None	None	Normal	Normal

Fibs = fibrillation potentials; **PSW** = positive sharp waves; **Fascics** = fasciculation potentials; **MUP** = motor unit potential morphology

EMG Interpretation

Findings consistent with acute left L5 radiculopathy.

Active denervation potentials (fibrillations and positive sharp waves) are present in muscles innervated by the left L5 nerve root (tibialis anterior, extensor hallucis longus, peroneus longus) and in the left L4-L5 paraspinal muscles. Muscles innervated by L4, S1, and S2 roots are normal, localizing the lesion to L5.

Motor unit potential morphology is normal, indicating acute (not chronic) denervation. This is consistent with the onset of symptoms approximately 4 weeks prior to study.

Nerve conduction studies are normal, which is expected in radiculopathy (the lesion is proximal to the dorsal root ganglion).

Electrodiagnostic impression:

1. Acute left L5 radiculopathy — active denervation, consistent with MRI findings of L4-L5 disc herniation.
2. No evidence of peripheral neuropathy.
3. No involvement of L4 or S1 myotomes.

Report Electronically Signed: Dr. Sarah Williams, MD — Neurology Dr. Laura Bennett, MD — Radiology Dr. Rachel Kim, MD — Cardiology

Report Date: January 9, 2024